

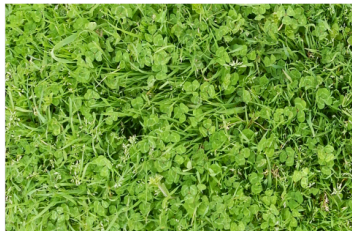
AIMS DTU Research Intern 2026

Neuro Symbolic AI

Preview

Neuro-Symbolic AI bridges the gap between human intuition and structured rules by fusing two completely different computing approaches:

1. The Neural Side (The "Gut Instinct"): Handles raw perception, pattern recognition, and statistical probabilities (e.g., recognizing a handwritten digit or a face).
2. The Symbolic Side (The "Logic Engine"): Handles abstract reasoning, explicit rules, and step-by-step logic (e.g., knowing that if $A > B$ and $B > C$, then $A > C$).



Your task is to predict biomass using the given below data through the usage of Logical Tensor Networks. The dataset comprises of the following:

1. Train.csv: Contains the training metadata and biomass targets for each pasture image. Each row represents one biomass component (target_name) associated with an image, along with additional information such as sampling date, pasture species, NDVI, height measurements, and the corresponding target biomass value in grams.
2. Images.zip: Contains the actual pasture images (.jpg) used for training the model. The image files are referenced in train.csv through the image_path column.

Your Assignment

You are expected to:

1. Integrate image features and auxiliary tabular metadata through a shared feature representation for joint learning and multi-target prediction.

2. Build a multi-modal neural network that grounds your 5 target pasture metrics as continuous, regression predicates.

3. Create a soft rule-based loss layer that helps the model make more consistent predictions.

The dataset link for the data: [Dataset](#)

Candidates are expected to make independent design decisions, evaluate their results critically, and clearly acknowledge the limitations or possible inaccuracies in their conclusions.

Evaluation

Evaluate your neuro-symbolic approach using:

- Biomass prediction accuracy for all 5 target metrics
- Quality of the learned shared feature representation
- Effectiveness of the soft rule-based loss layer in enforcing consistency
- Comparative analysis with baseline approaches (neural only, symbolic only)
- Interpretability of the learned rules and their semantic meaningfulness
- Critical assessment of your design choices and their limitations

Deliverables

1. A technical report (4-6 pages) covering your experimental setup, comparative analysis, results, interpretation of fuzzy logics, limitations, and directions for future work.

2. A GitHub repository with clean, reproducible code for LTN training, custom metric and predicate logic, and the feature extraction and fusion pipeline.